

DRUG TOXICITIES

There are 5 that you need to know!

1. LITHIUM

- Anti-mania drug
- Used for bipolar
- Not used for depression but for mania
- **Therapeutic level: 0.6 – 1.2**
- **Toxic level: 2.0 or greater**
- There is a grey area (1.2 – 2.0) → no books agree to what is going on with lithium at that area

2. LANOXIN/DIGOXIN

- Treats AFIB and CHF
- “DIG” = AFIB → abc**D**
- **Therapeutic level: 1.0 – 2.0**
- **Toxic level: 2.0 or greater**
- Therefore, 2.0 can be therapeutic or toxic → **If it gives you 2.0 you answer it is toxic rather than therapeutic (to be safe)**

3. AMINOFILIN

- Airway antispasmodic
- Technically not a bronchodilator → it does not stimulate the B2 agonists to bronchodilate, it just relaxes a spasm
- The airway dilates since it was narrowed but it is not a bronchodilator
 - o For instance, acute asthma attack → bronchodilator may not work initially due to spasm; therefore, use antispasmodic (aminofilin) to melt spasm then give bronchodilator
- **Therapeutic level: 10 – 20**
- **Under 10 is non therapeutic → need more or are they taking it?**
- **Toxic level: 20 or greater**

4. DILANTIN

- Used for seizures
- **Therapeutic level: 10 – 20**
- **Toxic level: 20 or greater**
- **Therefore, 20 is toxic**

5. BILIRUBIN

- Not a drug; byproduct of breakdown of RBCs
- Only tested in newborns on NCLEX
- Newborns have high bilirubin because breaking down mom's RBCs
- **Therapeutic level: Elevated level which is 10 – 20**
- **9.9 and less, is normal for a newborn**
- **Toxic level: 20 or greater**
- **Question:** A child with what bilirubin and above do you think needs to come to the hospital?
 - o Usually about 14-15 where doctors start thinking about hospitalizing these kids

- 10-13 can be managed with sunlight
- **Kernicterus:** Bilirubin in the brain which usually occurs when level is around 20 because it causes ASEPTIC (no germs) MENINGITIS and ASEPTIC ENCEPHALITIS due to irritation of bilirubin
- **Jaundice:** Yellow colour due to bilirubin in the skin
- **Opisthotonos:** Position the baby assumes when they have kernicterus → HYPEREXTEND due to irritation of meninges with the bilirubin
 - Newborns are unbelievably flexible
 - When they extend, their heels will come and touch their ears
 - Rigid
 - **Question on NCLEX:** What position do you place an opisthotonic child?
 - Put them on their side
- **Physiologic Jaundice:** Bilirubin is abnormal at birth, over the next 2-3 days goes high on bilirubin.
- **Pathologic Jaundice:** Bilirubin is high at birth, yellow at birth.

Therefore, if they come out yellow, SOMETHING IS WRONG. If they turn yellow, it's NORMAL.

NOTE: Lithium and Lanoxin both start with L. Their toxic level is 2.
The rest → Dilantin, bilirubin, aminofolin are 20.

- Is unrealistically preoccupied with fears of being left to take care of himself or herself

DUMPING SYNDROME VERSUS HIATAL HERNIA

	HIATAL HERNIA	DUMPING SYNDROME
Description	Regurgitation of acid into the esophagus because the upper part of stomach herniates upwards through the diaphragm. ○ Gastric contents move in the wrong direction ○ Stomach empties at a normal rate ○ GOING THE WRONG WAY ON A ONE WAY STREET	Usually follows gastric surgery in which the gastric contents dump too quickly into the duodenum ○ Gastric contents move in the right direction ○ Stomach empties at the wrong rate (increased) ○ YOU'RE SPEEDING
Signs and Symptoms	- Plain GERD → Heartburn and indigestion - You can have GERD for many reasons but GERD makes hiatal hernia - Hiatal hernia is GERD if you lie down after you eat - If you have indigestion and heartburn, doesn't mean you have hiatal hernia - Examples: ○ Nurse gets up in morning, skips breakfast, passes meds, then at 1100 gets epigastric burning pain, indigestion → GERD, yes. HIATAL HERNIA, no because she didn't lie down and didn't eat. ○ Go home, eat dinner at 8pm, sits down watches tv, half hour later eats cereal, 20 mins later eats chips, 20 mins later eats, then they get tired,	- Easiest way to remember is to take what you already know and combine it to equal dumping syndrome - Talk about DRUNK ○ Staggering gait ○ Slurred speech ○ Impaired judgment ○ Delayed reaction time ○ Labile emotions ○ BECAUSE with dumping syndrome → decreased cerebral perfusion - Talk about SHOCK ○ Hypotension ○ Tachycardia ○ Tachypnea ○ Pale, cold, clammy skin - DRUNK + SHOCK → HYPOGLYCEMIA - Then add ACUTE ABDOMINAL DISTRESS ○ Cramping

	they go to bed, half hour later → burning, epigastric pain, indigestion → HIATAL HERNIA, yes, because laid down right after they ate.	○ Pain ○ Guarding ○ Borborygmi ○ Tenderness ○ Diarrhea ○ Bloating ○ Distention - THEREFORE, S&S OF DUMPING SYNDROME: HYPOGLYCEMIA (DRUNK + SHOCK) + ACUTE ABDOMINAL DISTRESS
Treatment To change the way the stomach empties, you can: ○ Play with HOB ○ Play around with water content of meal ○ Play around with carbohydrate content of meal (carbs go through fast)	- you want stomach to empty FASTER because if it's empty, it won't reflux - 1. High position – gravity empties stomach faster 2. Liquid meal will go through the stomach faster therefore, increased liquids 3. High carbohydrate content “In HIGH-atal hernia, everything needs to be HIGH”	- you want the stomach to empty SLOWER 1. Low position – head flat → they should eat supine, turn to side, and eat on the side with head down 2. Low liquids – get fluids between meals (1 or 2 hrs before or after meals) 3. Low carbohydrate content “When everything is LOW, the stomach empties SLOW”
	What about protein? Low protein in hiatal hernia and high protein in dumping syndrome; whatever carbs is, protein is the opposite.	

ELECTROLYTES

You need to memorize three sentences:

1. Kalemia's (potassium) do the SAME AS the prefix except for heart rate and urine output.

- Prefix meaning HYPO or HYPER
- Hyperkalemia = Low urine output, low HR
- Hypokalemia = High urine output, high HR

DO NOT MEMORIZE THE LISTS but here are examples:

	HYPER KALEMIA	HYPO KALEMIA
Brain	Agitation, restlessness, Irritability, decreased inhibitions, loud	Lethargy, coma
Lungs	Tachypnea	Bradypnea
Heart	Bradycardia T waves are peaked and tall ST will be elevated ONLY THE HEART RATE IS OPPOSITE EVERYTHING ELSE IS SAME AS PREFIX	Tachycardia
Bowel	Borborygmi, diarrhea	Constipation, ileus, dec bowel sounds
Muscle	Spasticity, inc tone, 3+ or 4+ reflexes	Flaccid, 1+ reflexes
Kidneys	Oliguria/anuria	Polyuria

Question: Your patient has hyperkalemia, SATA:

THOUGHT PROCESS: Draw arrows (**v**) and (**^**) and draw exceptions (*****)

- Adynamic ileus (**v**)
- Obtunded 1+ reflexes (**v**)
- Clonus (irritability) (**^**)
- U wave (goes down) (**v**)
- Depressed ST (**v**)
- Polyuria (*****)
- Bradycardia (*****)

If you don't know what something is on SATA, don't pick it!

2. **Calcemia's do the OPPOSITE of the prefix.**

- a. HYPERcalcemia = everything goes down
- b. HYPOcalcemia = everything goes up
 - i. **CHvostek's sign** = Cheek → you touch their cheek → spasm which is a sign of neuromuscular irritability associated with a low calcium
 - ii. **Trousseau's sign** = spasm of hand when you put BP cuff on

3. **Magneemia's do the OPPOSITE of the prefix.**

Can it be a tie, that a symptom be caused by potassium, calcemia, magnesium?

- 1. Probably not magnesium because it's not a major player
- 2. If it is a skeletal muscle or nerve → blame it on Calcium
- 3. For everything else → blame it on Potassium

Question: Your patient has diarrhea. What caused it?

- A. Hyperkalemia (possibility)
- B. Hypokalemia (NO)
- C. Hypocalcemia (possibility)
- D. Hypomagneemia (possibility)

THOUGHT PROCESS: This symptom is UP (^)

Three possibilities

Rule out magnesium

Not a sk muscle or nerve

Blame it on potassium → **A**

Question: Your patient has tetany. What caused it?

- A. Hyperkalemia (possibility)
- B. Hypokalemia (NO)
- C. Hypocalcemia (possibility)
- D. Hypomagneemia (possibility)

THOUGHT PROCESS: This symptom is UP (^)

Three possibilities

Rule out magnesium

It IS a sk muscle or nerve

Blame it calcium → **C**

Question: Your patient has tetany. What caused it?

- A. Hyperkalemia (possibility)
- B. Hypokalemia (NO)
- C. Hypercalcemia (NO)
- D. Hypomagneemia (possibility)

THOUGHT PROCESS: This symptom is UP (^)

Three possibilities

Rule out HYPERcalcemia because it is the OPPOSITE

Don't just think calcium = muscles/nerves

Don't just think potassium = heart

Therefore, it is a tie breaker between magnesium and potassium → Pick potassium → A

SODIUM'S

D
HypEnatremia
H
Y
D
R
A
T
I
O
N

HypOnatremia
V
E
R
L
O
A
D

HypErnatremia → See letter **E** → dEhydration
HypOnatremia → See the **O** → Overload

Question: A student nurse runs to RN → I just ran a whole litre of sodium in 10 minutes, I forgot to close the clamp. What electrolyte imbalance would you expect to see? Hyponatremia because of fluid overload.

Question: Which patient is put on a fluid restriction and Lasix? Hyponatremia.

Question: Which one is given lots of fluids? Hypernatremia.

Question: Who has hot, flushed, dry skin? Hypernatremia.

Question: In addition to HYPERKALEMIA in DKA, what other electrolyte imbalance is possible? Patient is Dehydrated → Hypernatremia

Question: What nursing diagnosis would be essential for hyponatremia? Fluid volume excess

Therefore:

- For SIADH → overload → Hyponatremia
- For DI → dehydration → Hypernatremia
- For HHNK → dehydration → Hypernatremia

Other Notes:

- ✓ **The earliest sign of any electrolyte imbalance = NUMBNESS AND TINGLING**
 - **PARESTHESIA** = numbness and tingling
- ✓ **Circumoral paresthesia** = numbness and tingling of the lips
- ✓ **All electrolyte imbalances cause muscle weakness**
 - **PARESIS:** Muscle weakness

TREATMENT

- Only tests potassium*

1. **Never push potassium IV**
2. **Not more than 40 of K per litre of IV fluid**
 - a. If more, call MD and clarify

These are ways to lower potassium:

High potassium is bad because it stops your heart → cardiac arrest

Therefore, HYPERkalemia is the worst electrolyte imbalance!!!

3. **Give D5W with regular Insulin** → drives potassium into the cell and out of the blood
 - a. Fastest way to lower potassium
 - b. It is the potassium in the blood that will kill you, not the potassium in the cells
 - c. Does D5W get rid of the extra potassium? NO. It just shifts it into the cells.
 - d. Does it solve the problem? NO.
 - e. Why do we do it? To save their life!!!
 - f. Potassium will leak right back into the blood after a couple of hours
 - g. Upside: FAST AND QUICK
 - h. Downside: TEMPORARY
4. **Kayexelate**
 - a. Goes into your but (enema) or oral
 - b. It is full of SODIUM

- c. It releases sodium into your blood
- d. But to maintain negative equilibrium → potassium is kicked out of the blood into the gut
- e. Trades SODIUM for POTASSIUM
- f. Defecate K
- g. Blood starts out hyperkalemic and ends up with hypernatremia → DEHYDRATION
 - i. Therefore, cure the hypernatremia with fluids
- h. Upside: Get rid of the excess potassium out of the body → doesn't reoccur
- i. Downside: It takes a long time!!! Hours, and you may not live that long

What electrolyte does Kayexelate work with? "KAY"

Does it make potassium enter the cell or exit the body? "EXit"

Does it do it fast or slow? Early or late? "LATE"

KAYEXITLATE. Kay exits late.

D5W with Regular Insulin = opposite. Kay enters fast. Potassium enters the cells fast.

THEREFORE, GIVE D5W AND REGULAR INSULIN + KAYEXELATE SIMULTANEOUSLY!

ENDOCRINE OVERVIEW

Thyroids and Adrenals and you're good to go!

HYPERTHYROIDISM

Turn "THYROIDISM" into "METABOLISM" → HyperMETABOLISM

SIGNS & SYMPTOMS:

- Weight loss
- Hypertension
- Irritable
- Heat intolerance
- Cold tolerance
- Exophthalmos (bulging eyes)
- Tachycardia

Grave's Disease → Hyperthyroidism → "You're going to run yourself into the grave"

TREATMENT

1. Radioactive Iodine
 - a. Patient should be by themselves for 24 hours (private room)
 - b. After that, they have to be very careful with urine → flush three times; if they spill urine on the ground → call hazardous team NOT housekeeping
 - c. Risk to nurse → URINE
2. PTU (propyl thyo uracil)
 - "PUTS THYROID UNDER"
 - Must bring down thyroid hormones
 - Primary use is for cancer but one of its special uses is for thyroids
 - Watch WBC due to immunosuppression
3. Surgical Removal → Thyroidectomy (remove part or all of the thyroid)
 - Pay attention to if it is a total or subtotal thyroidectomy; if you treat all the same, you will get it wrong!!!

TOTAL THYROIDECTOMY

- Life long hormone replacement
- You are at risk for HYPOCALCEMIA because it's almost impossible to spare the
- Parathyroid Gland → Low PTH, low Calcium
- S&S -- calcium do the opposite of the prefix

PARTIAL THYROIDECTOMY

- May be on hormones for a bit but not life long
- No hypocalcemia with this; however, they are at risk for
 - **THYROID STORM or THYROTOXICOSIS** (total thyroidectomies never are at risk for this) → MEDICAL EMERGENCY → BRAIN DAMAGE → PERMANENT
 - S&S (**FOUR** REALLY BAD THINGS)
 1. SUPER high temperatures of 105F and above
 2. Extremely high BP → Stroke category → 210/180 (for instance)
 3. Severe tachycardia → 180's - 200
 4. Psychotically delirious

- TREATMENT of Thyrotoxicosis:
 - Get temperature DOWN
 - FIRST: Ice packs
 - BEST: Cooling blanket
- Get oxygen UP
 - O2 per mask @ 10L
 - How can you keep an oxygen mask on this delirious patient? This is hard.
- OXYGEN FIRST THEN TEMPERATURE but always pick STAY WITH PATIENT
 - They will come out of it themselves or they will die
 - You do not want to medicate
 - For instance, tylenol won't work because the hypothalamus is not working.
 - This is self limiting condition
 - All we do is SPARE THE BRAIN until they come out of it
 - Two staff for ONE patient (2:1)

POST OP RISKS

In the **first 12 hours**, no matter if it's total/subtotal,

1. Airway
 - Edema → presses on airway
2. Hemorrhage
 - Endocrine gland has lots of BV

After 12 hours...

- A. Post op risks 12-48 hours after a **TOTAL** → **TETANY** due to HYPOcalcemia
- B. Post op risks 12-48 hours after a **SUBTOTAL** → **STORM**

T's with T. S's with S.

After 48 hours...

Biggest risk is **infection**.

Never pick infection before 72 hours WITH anything!!!

HYPOTHYROIDISM → Hypo Metabolism

SIGNS & SYMPTOMS

- Flat, boring, dull
- Cold intolerance
- Heat tolerance
- Bradycardia
- Slow processing
- Bradypnea

MYXEDEMA COMA is the disease.

TREATMENT

1. Not enough hormone therefore, GIVE THEM THYROID HORMONES (Synthroid/levothyroxine)
2. Do not sedate these people because they're already super slow → coma
 - a. QUESTION NPO before surgery because then they can't take their meds!!! If they don't have their hormones → anaesthetic in sx can kill them
 - b. NEVER HOLD SYNTHROID without expressed orders to do so

ADRENAL CORTEX

Diseases start with either A or C (same as Adrenal Cortex)!

ADDISON'S DISEASE → Under secretion of the adrenal cortex

SIGNS & SYMPTOMS

Two things you need to know...

1. Hyperpigmented
 - a. Very tanned
 - b. They look healthy
2. They do not adapt to stress
 - . When you are under stress there's a threat to your brain
 - a. Purpose of stress response → raise glucose and raise BP
 - b. THEREFORE → Hypoglycemia and hypotension → shock

TREATMENT

1. Steroids
 - a. End in -sone
 - b. "ADDISONS YOU ADD A SONE"

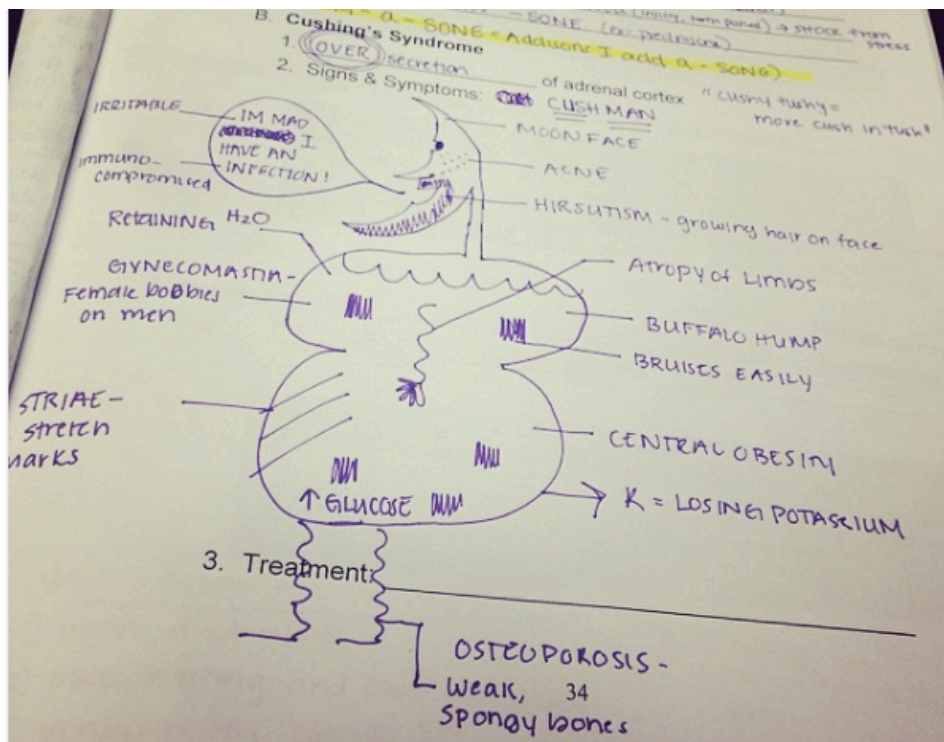
CUSHING'S SYNDROME → Over secretion of the adrenal cortex

If you have a "cushy" bank account
If you have a "cushy" chair
If you have a "cushy" tushy
YOU HAVE MORE...

SIGNS & SYMPTOMS → YOU HAVE TO KNOW THIS

Memorize this list and you will get a 2 for 1 → S&S of Cushing's AND side effects of CORTICOSTEROIDS (-sone)!

Draw a picture of little man is called CUSH-MAN.



1. Moon face
2. Acne
3. Hirsutism
4. Retaining water
5. Gynecomastia
6. Increased bruising
7. Muscle atrophy
8. Central obesity
9. Losing K
10. Osteoporosis
11. Striae
12. Increased glucose → Need accuchecks! Even if you're not diabetic.
13. Irritable

TREATMENT

You have too much → You need to cut it out → ADRENALECTOMY

- **Bilateral adrenalectomy:** Problem is you can induce Addison's disease
 - Then you have to take corticosteroids
 - Then you end up looking like Cushman again!!!
 - Takes a year or two → normal

*This is the same as hyperthyroidism: You get a thyroidectomy → take thyroid hormones → signs and symptoms look like grave's disease